

From the full stack to Spatial only

We started with the full stack: a CLIP spatial tower, an RGPA forensic tower, gated fusion of their logits, and MBE at eval. Then we removed each piece on the same 5,000-image WildFake val slice. Keep a piece only if it raises official score ($0.5 \times \text{AUC}_{\text{clean}} + 0.5 \times \text{mean of 14 degradations}$). MBE, RGPA, and fusion all lose. We submit Spatial only - OpenCLIP ViT-H/14, taken at minimum val_loss, not at the later AUC peak.

0.9924 Spatial official (submitted)	-0.0024 Fusion - Spatial	-0.0023 MBE - Spatial	0.9064 RGPA official
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Why start with two towers and enhancement

Sharing wipes high-frequency traces first. The spatial tower reads semantics; RGPA was meant to catch local generation traces; MBE was meant to restore them before scoring. We built all three, then took them off one at a time. Official degradations at train time stay - that is the robustness recipe. MBE, RGPA, and the gate do not.

Submitted Spatial

OpenCLIP ViT-H/14 on the 5,000-image WildFake val slice (50/50). Acc / F1 / precision / recall use threshold 0.5. Robust is the mean of 14 degradations. Official is $0.5 \times \text{clean} + 0.5 \times \text{robust}$. Worst condition is Noise 10%.

0.9924 Official AUC	0.9931 Official AP	0.9315 Official Acc @ 0.5	0.0438 Official EER
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	AUC	AP	Acc	F1	Prec.	Rec.	EER	ECE	Brier
Clean	0.9966	0.9968	0.9392	0.9354	0.9973	0.8808	0.0298	0.0561	0.0492
Robust	0.9882	0.9893	0.9238	0.9183	0.9890	0.8571	0.0579	0.0665	0.0646
Official	0.9924	0.9931	0.9315	0.9269	0.9932	0.8689	0.0438	0.0613	0.0569

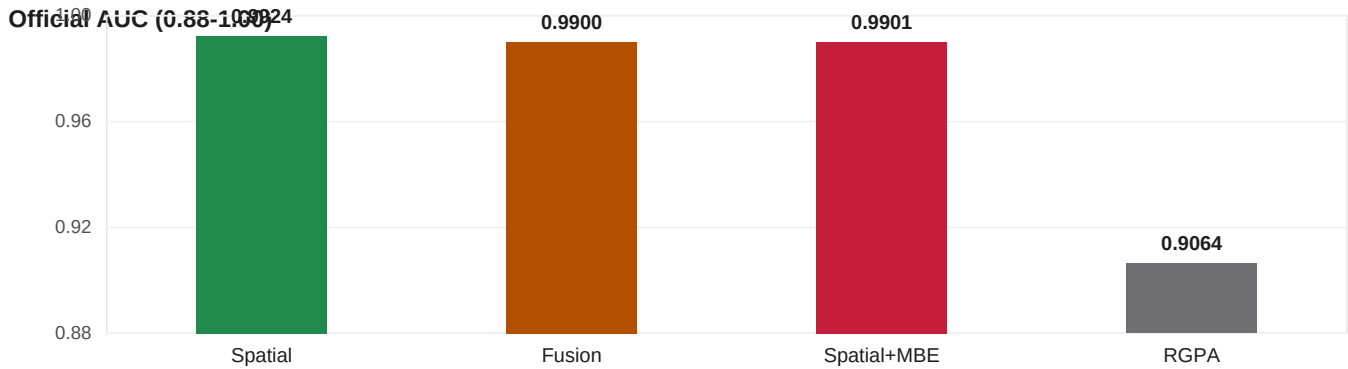
At 0.5 the model is conservative: precision 0.997 / recall 0.881 on clean.

Condition	AUC	AP	Acc	F1	EER
Clean	0.9966	0.9968	0.9392	0.9354	0.0298
JPEG 90	0.9964	0.9966	0.9356	0.9314	0.0304
JPEG 70	0.9955	0.9958	0.9258	0.9200	0.0360
JPEG 50	0.9938	0.9945	0.9306	0.9255	0.0408
JPEG 30	0.9902	0.9913	0.9260	0.9204	0.0546
Blur 0.5	0.9958	0.9961	0.9370	0.9329	0.0332
Blur 1.0	0.9871	0.9882	0.9242	0.9192	0.0632
Blur 2.0	0.9766	0.9793	0.9056	0.8977	0.0904
Resize 1/2	0.9939	0.9944	0.9352	0.9309	0.0418
Resize 1/4	0.9871	0.9887	0.9228	0.9171	0.0634
Noise 2%	0.9811	0.9823	0.9186	0.9139	0.0836
Noise 5%	0.9816	0.9829	0.9080	0.9004	0.0796
Noise 10% (worst)	0.9707	0.9738	0.8998	0.8917	0.1010
Color	0.9938	0.9942	0.9374	0.9335	0.0414
Crop 80%	0.9914	0.9920	0.9264	0.9211	0.0508

Official 15-condition score

Same 5k WildFake val slice and Spatial weights for every bar. Fusion is frozen towers plus a gate. MBE is an

eval-only preprocess in front of Spatial. This is the number that decided what to submit.



Spatial 0.9924 | Fusion 0.9900 | Spatial+MBE 0.9901 | RGPA 0.9064

By degradation: Spatial vs RGPA vs Fusion vs MBE

Blur / Resize / Noise average the official sub-parameters. Deltas are versus Spatial - negative means the add-on did not help. Fusion fails most on noise (the residual is gone, but the gate still mixes in about 25% RGPA). MBE fails most on blur and crop.

Condition	Spatial	RGPA	Fusion	d Fusion	Spatial+MBE	d MBE
Clean	0.9966	0.9260	0.9956	-0.0010	0.9950	-0.0016
JPEG 90	0.9964	0.9034	0.9954	-0.0010	0.9947	-0.0017
JPEG 70	0.9955	0.9062	0.9943	-0.0012	0.9937	-0.0018
JPEG 50	0.9938	0.9134	0.9920	-0.0018	0.9925	-0.0013
JPEG 30	0.9902	0.9226	0.9880	-0.0022	0.9896	-0.0006
Blur	0.9865	0.8803	0.9861	-0.0004	0.9782	-0.0083
Resize	0.9905	0.8809	0.9903	-0.0002	0.9876	-0.0029
Noise	0.9778	0.8433	0.9627	-0.0151	0.9783	+0.0005
Color	0.9938	0.9214	0.9928	-0.0010	0.9913	-0.0025
Crop	0.9914	0.9167	0.9915	+0.0001	0.9845	-0.0069
Official	0.9924	0.9064	0.9900	-0.0024	0.9901	-0.0023

RGPA is 0.07-0.13 lower on every family. Fusion drops most on noise; MBE drops most on blur and crop.

Train-loop val (not the official 15-grid)

Fit on the training distribution - not the shipping decision. On SID the two towers look interchangeable; on WildFake, RGPA collapses and Spatial does not. Train-loop robust uses one random official transform per image, not the mean of 15 conditions. Do not compare these with the 5k official scores above.

Run	Data	clean AUC	robust AUC	Notes
SID Spatial	SID val	1.0000	1.0000	epoch 3; no 15-grid eval
SID RGPA	SID val	0.9999	0.9993	epoch 27; 15-grid official 0.9995
WildFake Spatial	WildFake val	0.9988	0.9966	submitted; min val_loss; last 2 blocks
WildFake Probe	WildFake val	0.9989	0.9966	backbone frozen; best at step 1000
WildFake RGPA	WildFake val	0.9649	0.9363	weaker than CLIP-H after domain shift

Unfreeze depth (WildFake Spatial)

Two freeze settings: last 2 of 32 ViT-H blocks unfrozen, and a linear probe with the backbone frozen. No unfreeze-1 / 4 / 8. The probe starts from the submitted Spatial weights and freezes the tower. Unfreeze-2 is what we keep; the last two blocks are already enough.

Setting	Trainable	Init	step	clean	robust	loss
Unfreeze 2 (submitted)	last 2 + ln_post + proj + head	SID Spatial	10000	0.9988	0.9966	0.0487
Unfreeze 2 (AUC peak)	same	SID Spatial	12000	0.9990	0.9964	0.1132
Linear probe	proj + head only	submitted Spatial	1000	0.9989	0.9966	0.0465

Training tracked *val_auc*, so the AUC peak is at step 12000. We submit the min *val_loss* snapshot instead (step 10000, loss 0.0487). By 12000, *val_loss* is 0.1132 - the AUC peak is already overfitting. The linear probe starts from that snapshot. SID Spatial also used unfreeze 2.

Unfreeze 2 - val every 1000 steps

step	epoch frac	val clean	val robust	val loss	train loss
1000	0.04	0.9907	0.9727	0.1540	0.4424
2000	0.07	0.9956	0.9862	0.1021	0.2702
3000	0.11	0.9969	0.9908	0.0841	0.2042
4000	0.14	0.9978	0.9934	0.0727	0.1672
5000	0.18	0.9981	0.9947	0.0630	0.1434
6000	0.21	0.9983	0.9953	0.0593	0.1269
7000	0.25	0.9984	0.9955	0.0569	0.1147
8000	0.28	0.9986	0.9960	0.0495	0.1053
9000	0.32	0.9988	0.9966	0.0508	0.0978
10000 (submitted)	0.35	0.9988	0.9966	0.0487	0.0912
11000	0.39	0.9988	0.9970	0.0507	0.0859
12000	0.42	0.9990	0.9964	0.1132	0.0805
13000	0.46	0.9989	0.9960	0.1495	0.0750
14000	0.49	0.9987	0.9957	0.1751	0.0703
15000	0.53	0.9986	0.9952	0.1786	0.0661
16000	0.56	0.9986	0.9954	0.1791	0.0624
17000	0.60	0.9982	0.9945	0.2271	0.0591
18000	0.63	0.9979	0.9945	0.2274	0.0563
19000	0.67	0.9975	0.9933	0.2212	0.0537
20000	0.70	0.9971	0.9926	0.2515	0.0513

After step 10000, train loss keeps falling while val loss blows up. That is why we submit min *val_loss*, not the later AUC peak.

Linear probe (backbone frozen)

Steps after freezing, not unfreeze-2 steps. *val_loss* stays near 0.047; AUC does not beat the unfreeze-2 peak.

probe step	val clean	val robust	val loss
1000 (best)	0.9989	0.9966	0.0465
2000	0.9988	0.9963	0.0481
3000	0.9988	0.9963	0.0489

CIFAKE smoke (not used for the decision)

First run only: prove train / eval works. Images are 32x32; upsampling to 224 does not restore high frequency, so these scores are not an upper bound. CLIP-H 15-grid: clean 0.9996, robust 0.9956, official 0.9976, worst condition resize 0.25 (0.9787).